

Pure Mathematics

Thursday 8 October 2026 · Student handout · Original examples · No solutions

Least elements & valid remainders

Keep the hypotheses visible

Well-ordering: every nonempty subset of $\mathbb{Z}_{\geq 0}$ has a least element.

Division: for integers $a \neq 0$ and b , write $b = aq + r$ with $0 \leq r < |a|$.

An integer set and a real set

Compare the two sets below. List four elements of S , identify its least element, and explain why well-ordering applies. Does T have a least element? Explain.

$S = \{m \in \mathbb{Z}_{\geq 0} : m > 17/4\}, \qquad T = \{x \in \mathbb{R} : x > 17/4\}$

S: elements, least element and reason

T: decision and reason

Check the equality and the interval

Find integer values of q, r in each case. Check both conditions, not just the equality.

Equality	q	r	Remainder check
$-29 = 6q + r$			$0 \leq r < 6$
$29 = (-6)q + r$			$0 \leq r < 6$

For the first case, a student proposes $q = -4, r = -5$. Is the equality correct? Is this the required division-algorithm answer?

Two separate checks

Remove the remainder restriction: give two pairs that satisfy $-29 = 6q + r$.

Two pairs

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Why the proof works

A definition you can use

$d \mid n$ means $n = dk$ for some $k \in \mathbb{Z}$. A proof of divisibility must identify an **integer witness**.

What makes the remainder unique?

Suppose $-29 = 6q + r = 6q' + r'$ with integer variables and $0 \leq r, r' < 6$. Prove the two representations agree. Use both the equality and the bounds.

Subtract the two equalities

Use the bounds on the remainders

Conclude: why must the difference vanish?

Prove a claim; test a stronger one

Assume $4 \mid u$ and $4 \mid v$. Prove $4 \mid (3u - 2v)$. Your last line should say why the witness is an integer.

Definition $\rightarrow$ algebra $\rightarrow$ conclusion

A student says that $8 \mid (3u - 2v)$ follows too. Decide by proof or counterexample. Check the original assumptions.

Decision and supporting evidence

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Common divisors & a Bézout certificate

What gcd means

$\gcd(a, b)$ is the **greatest positive common divisor** (when a, b are not both zero). A linear-combination identity alone does not prove a number is the gcd.

Start with the definition

List the positive common divisors of 18 and 30, then identify their greatest common divisor.

Common divisors and gcd

Use an identity as a certificate

You are given $6 = 2 \cdot 18 - 30$. Let c be any common divisor of 18 and 30. Explain why $c \mid 6$, then why the identity certifies $\gcd(18, 30) = 6$. No coefficient-finding algorithm is needed.

Every common divisor divides 6

The additional fact needed to certify the gcd

Can integers x, y satisfy $18x + 30y = 9$? Explain.

Decision and reason

Check your understanding

Without notes, complete $22 = (-5)q + r$ with $0 \leq r < 5$.

$q =$ _____ $r =$ _____ Equality check:

What two facts certify a positive integer g as $\gcd(a, b)$? Name one hypothesis required for well-ordering.

Your explanation

